

SOHAIL DASGUPTA

Curriculum Vitae



Postdoctoral Research Fellow, University of Birmingham |

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RESEARCH INTERESTS

Quantum Simulation, Ultracold Atoms and Molecules, Correlated quantum matter, Quantum Dynamics

ACADEMIC POSITIONS

2025 - Postdoctoral Research Fellow at the Theoretical Physics Division, University of Birmingham.

EDUCATION

2019 - 2025 PhD (Physics) at *Rice University* (Awarded on: Aug 31, 2025.)

Thesis Advisor: Dr. Kaden Hazzard

Title: Collective quantum phenomena in strongly correlated multi-level systems

2014 - 2019 BS-MS (Physics) at *Indian Institute of Science Education and Research (IISER) Kolkata*

AWARDS

APS DAMOP Student Participation Award (*Portland, Oregon*) 2025

APS DAMOP Student Participation Award (*Fort Worth, Texas*) 2024

INSPIRE (SHE) Fellow of the Dept. of Science and Technology, Govt. of India 2014.

PUBLICATIONS

Journal Articles

- Kim, D., Dasgupta, S., Ma, X., Park, J.-M., Wei, H.-T., Li, X., Luo, L., Doumani, J., Yang, W., Cheng, D., Kim, R. H. J., Everitt, H. O., Kimura, S., Nojiri, H., Wang, J., Cao, S., Bamba, M., Hazzard, K. R. A. & Kono, J. Observation of the magnonic Dicke superradiant phase transition. *Science Advances* **11**, eadt1691 (Apr. 2025).
- Dasgupta, S., Feng, C., Gadway, B., Scalettar, R. T. & Hazzard, K. R. A. Finite-temperature quantum matter with Rydberg or molecule synthetic dimensions. *Phys. Rev. A* **109**, 063322 (June 2024).
- Pasqualetti, G., Bettermann, O., Darkwah Oppong, N., Ibarra-García-Padilla, E., Dasgupta, S., Scalettar, R. T., Hazzard, K. R. A., Bloch, I. & Fölling, S. Equation of State and Thermometry of the 2D $SU(N)$ Fermi-Hubbard Model. *Phys. Rev. Lett.* **132**, 083401, **Featured in Physics, Editor's suggestion** (Feb. 2024).
- Kanungo, S., Whalen, J. D., Lu, Y., Yuan, M., Dasgupta, S., Dunning, F. B., Hazzard, K. R. A. & Killian, T. C. Realizing topological edge states with Rydberg-atom synthetic dimensions. *Nature communications* **13**, 972 (Feb. 2022).
- Ibarra-García-Padilla, E., Dasgupta, S., Wei, H.-T., Taie, S., Takahashi, Y., Scalettar, R. T. & Hazzard, K. R. A. Universal thermodynamics of an $SU(N)$ Fermi-Hubbard model. *Phys. Rev. A* **104**, 043316, **Editor's suggestion** (Oct. 2021).

Preprints

- Raghuram, A. P., Blondell, F. M., Mortlock, J. M., Maddox, B. P., Dasgupta, S., Middleton-Spencer, H. A. J., Hazzard, K. R. A., Price, H. M., Gregory, P. D. & Cornish, S. L. Probing topological edge states in a molecular synthetic dimension. *arXiv:2604.00745* (2026).

2. Mongkolkiattichai, J., Liu, L., Dasgupta, S., Hazzard, K. R. A. & Schauss, P. Quantum gas microscopy of three-flavor Hubbard systems. [arXiv:2503.05687](https://arxiv.org/abs/2503.05687) [In line for copy-editing Science] (2025).

TEACHING

Instructor

Research Proposal and Writing Seminar *Rice University* Summer 2022

Teaching Assistant

Undergrad Classical Mechanics	<i>Rice University</i>	Fall 2022
Grad Classical Electrodynamics	<i>Rice University</i>	Spring 2021
Mechanics, Phys 101	<i>Rice University</i>	Fall 2020
Mathematical Methods I	<i>Indian Institue of Sciene Education and Research Kolkata</i>	Spring 2019
Condensed Matter Laboratory	<i>Indian Institute of Science Education and Research Kolkata</i>	Fall 2018

CONFERENCES AND SEMINARS

Invited Talks

Theory Seminar	<i>University of Birmingham, UK</i>	Mar 2026
Trinity University Weekly Physics Seminar	<i>Trinity University, San Antonio</i>	Sep 2024
Rice Quantum Initiative Seminar	<i>Rice University</i>	Mar 2023
UC Davis Condensed Matter Seminar	<i>UC Davis</i>	Feb 2023
Kalyani University Physics seminar	<i>Kalyani University, India</i>	Jan 2023

Contributed Talks

APS DAMOP Meeting	<i>Portland</i>	Jun 2025
Space City Symposium	<i>University of Houston</i>	Apr 2025
eQMA Workshop	<i>Rice University</i>	Oct 2024
APS DAMOP Meeting	<i>Fort Worth</i>	Jun 2024
Space City Symposium	<i>Rice University</i>	Mar 2024
Smalley-Curl Institute Colloquium	<i>Rice University</i>	Aug 2023
APS March Meeting	<i>Las Vegas</i>	Mar 2023
Texas Section of APS Meeting	<i>Rice University</i>	Sep 2022
APS DAMOP Meeting	<i>Orlando</i>	June 2022
APS March Meeting	<i>Chicago</i>	Mar 2022
APS DAMOP Meeting	<i>Online</i>	June 2021
APS March Meeting	<i>Online</i>	Mar 2021

Contributed Posters

ICTP-IBM School on Quantum Computing and Simulation	<i>Trieste, Italy</i>	Apr 2026
Winter School on Ultracold Quantum Many-Body Systems	<i>Benasque, Spain</i>	Feb 2026
QQQS Workshop	<i>Oxford University</i>	Jan 2026
ITAMP Winter School	<i>Biosphere 2</i>	Feb 2025
Gordon Research Conference Quantum Science	<i>Stonehill College</i>	July 2024
Quantum Vacuum Matter	<i>Rice University</i>	Oct 2023

LEADERSHIP

Leadership Positions

President	<i>Rice Physics and Astronomy Graduate Student Association</i>	2022-2023
Treasurer	<i>Rice Physics and Astronomy Graduate Student Association</i>	2021-2022
Journal Club Coordinator	<i>Rice Physics and Astronomy Graduate Student Association</i>	2020-2021

Mentorship Roles

Wellbeing Peer	<i>Rice University</i>	2022-Present
Graduate student mentor (2021-present)	<i>Rice University</i>	2021-present

TECHNICAL SKILLS

Programing Languages: Python, MATLAB, Mathematica, C, FORTRAN